

SABLE: Zero-Knowledge Biometric Authentication with Spatial Flash Liveness

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Abstract—Biometric authentication systems face a fundamental tension between usability and privacy: conventional approaches require centralized storage or cloud processing of irrevocable biometric data, creating high-value targets for breaches and surveillance. We present SABLE (Secure Attested Biometric Library for Edge), a system that performs biometric identity verification entirely through zero-knowledge proofs, ensuring that biometric data never leaves the user’s device. SABLE combines Halo2-based ZK circuits over quantized face embeddings with a novel spatial flash liveness protocol that detects presentation attacks by exploiting differential light reflectance on 3D facial geometry. The system operates fully offline via peer-to-peer communication and supports selective credential disclosure through BBS+ signatures. On commodity hardware (Apple Silicon), SABLE generates combined face-match and liveness proofs — with the entire match, template-commitment binding, and liveness check performed in-circuit — in approximately 1 second with 2.08 KB proof size, and verifies them in approximately 1.8 ms — well within typical interactive latency budgets. To our knowledge, SABLE is the first open-source system to combine true zero-knowledge biometric proofs, transparent setup, offline peer-to-peer verification, and screen-based liveness detection without specialized hardware.

Index Terms—zero-knowledge proofs, biometric authentication, privacy, liveness detection, Halo2

I. INTRODUCTION

Biometric authentication offers strong identity assurance — a person’s face or fingerprint is difficult to forge and impossible to forget. Yet every major deployment model compromises on privacy. Centralized biometric databases create catastrophic breach risks: the 2015 U.S. Office of Personnel Management breach exposed 5.6 million fingerprint records, and India’s Aadhaar system has faced repeated data exposure incidents. Once biometric data is compromised, it cannot be revoked or reissued.

Multi-party computation (MPC) offers one approach, splitting biometric templates across servers so no single entity holds the complete data. However, this still requires cloud infrastructure, trust in the operating parties, and constant network connectivity. Blockchain-based identity systems achieve uniqueness through on-chain verification but introduce internet dependencies, transaction costs, and lack inherent privacy-preserving biometric matching. Hardware-dependent solutions such as iris scanners achieve high accuracy but require specialized sensors, constraining deployment to controlled environments.

Contributions. We present SABLE, a system that resolves these trade-offs by combining several techniques into a coherent architecture:

- 1) **ZK biometric matching.** We construct Halo2 arithmetic circuits that prove a quantized face embedding matches an enrolled template within a Hamming distance threshold, without revealing the embedding. The circuit operates over 512-byte quantized feature vectors using Poseidon hashing and Pedersen commitments.
- 2) **Spatial flash liveness.** We integrate a screen-based liveness detection protocol inspired by Tang et al. [1] into the ZK proof. A commit-reveal nonce scheme generates unpredictable split-screen color flashes; differential reflectance analysis distinguishes 3D faces from flat presentation attacks. The liveness evidence is quantized into 16-bit delta fingerprints and verified within the same Halo2 circuit.
- 3) **Offline selective disclosure.** SABLE supports optional BBS+ signed Verifiable Credentials, enabling users to prove predicates (e.g., “age ≥ 18 ”) without revealing underlying data. The Pedersen commitment anchors the biometric proof to the credential, and verification requires no internet connection.
- 4) **Practical performance.** On Apple Silicon, combined proof generation takes ~ 250 ms with a 2.08 KB proof, and verification completes in ~ 1.8 ms — all within a transparent-setup (no trusted ceremony) proof system.

II. SYSTEM ARCHITECTURE

SABLE comprises three principal roles: the *user* (prover), the *ZK engine*, and the *verifier*. An optional *identity issuer* (e.g., a government authority) may provide BBS+ signed Verifiable Credentials.

A. Enrollment

During enrollment, the user captures a face image via a standard smartphone camera. A feature extraction model produces a 1024-dimensional floating-point embedding $\mathbf{e} \in \mathbb{R}^{1024}$, which is reduced and thermometer-encoded into a 512-byte vector $\mathbf{q} \in \{0, \dots, 255\}^{512}$ (Section III-A). The quantized embedding is hashed via Poseidon into a field element $h = \text{Poseidon}(\mathbf{q})$ and then committed using a Pedersen commitment $C = g^h \cdot h_r^s$ over BLS12-381, where s is a 256-bit random salt. The commitment C serves as the

user’s “biometric public key” — stored by the verifier but cryptographically hiding the underlying biometric data.

B. Authentication

To authenticate, the user recaptures their face and undergoes the spatial flash liveness protocol (Section IV). The ZK engine then generates a Halo2 proof attesting that: (1) the fresh embedding matches the enrolled template within a Hamming distance threshold, (2) the liveness challenge was correctly completed with the HKDF-derived parameters, and (3) the proof is bound to a fresh challenge nonce. The proof exposes four public inputs — face-match result, threshold, liveness result, and a packed challenge digest — while no biometric data crosses the device boundary.

C. Selective Disclosure

If the user holds a BBS+ signed Verifiable Credential from an identity issuer, they may selectively disclose credential attributes alongside the biometric proof. The Pedersen commitment acts as the binding anchor between the biometric data and the credential. The Halo2 proof simultaneously covers biometric match and credential predicates in a single composite proof, so the verifier learns only the disclosed predicates (e.g., “age ≥ 18 ”) and the biometric match result.

D. Communication

SABLE operates peer-to-peer over NFC, BLE, or WiFi Direct. Session keys are established via X25519 ECDH and traffic is encrypted with ChaCha20-Poly1305 AEAD. No internet connection or central server is required for verification.

III. ZK FACE VERIFICATION

The core ZK circuit proves that a fresh biometric capture matches an enrolled template without revealing either. We describe each stage of the pipeline.

A. Embedding Quantization

Face embeddings from neural feature extractors are 1024-dimensional vectors of 64-bit floats, reduced to a 512-component representation and normalized to $[-1, 1]$ for the matcher. We quantize each component to an unsigned 8-bit integer:

$$q_i = \text{clamp}(\lfloor (e_i + 1.0) \times 127.5 \rfloor, 0, 255) \quad (1)$$

This linear map to 256 levels is the *binary* baseline encoding. The deployed matcher instead uses an ordinal *thermometer* encoding of the same 8-bit-per-component budget (Section VI): after a per-dimension prescale into $[-1, 1]$, each component is quantized to one of nine levels and written as k contiguous low bits, so that the Hamming distance between two codes equals the $L1$ distance between their quantization levels. Section VI shows this ordinal encoding recovers accuracy the binary map discards, at no additional in-circuit cost. Either way the result is a 512-byte vector, which is bound by the Poseidon commitment (below) and matched by the Hamming-distance circuit.

B. Poseidon Commitment

The quantized embedding $\mathbf{q}$ is hashed using the Poseidon hash function [2], which is designed for efficient evaluation inside arithmetic circuits. We use Poseidon with width $t = 9$, rate $r = 8$, $R_F = 8$ full rounds, and $R_P = 56$ partial rounds over the BN254 scalar field. The output field element $h = \text{Poseidon}(\mathbf{q})$ is then used in a Pedersen commitment $C = g^h \cdot h_r^s$ over BLS12-381. The verification circuit additionally recomputes the commitment to the enrolled template in-circuit and exposes it as a public input, which the verifier checks against the value registered at enrollment. This binds the proof to the correct template: a prover cannot match against an embedding other than the one on file, even with an otherwise valid proof.

C. Hamming Distance Circuit

Rather than computing Euclidean distance (which requires expensive field multiplications), we measure similarity via Hamming distance over bit representations. Given the 512-byte encoded embeddings $\mathbf{p}$ and $\mathbf{r}$, the circuit expands them to 4096 bits and computes:

$$d_H(\mathbf{p}, \mathbf{r}) = \sum_{i=1}^{4096} (p_i \oplus r_i) \quad (2)$$

Under the deployed thermometer encoding this Hamming distance equals the $L1$ distance between the two templates’ per-component quantization levels (Section VI); the circuit also constrains each byte to be a valid thermometer code, so this equivalence cannot be subverted. The XOR and bit-count operations are implemented over a lookup-table-backed bit decomposition (2^{13} -entry tables in a $k = 16$ circuit), making this far more efficient than field arithmetic over floating-point representations. The distance is computed entirely in-circuit from the two embeddings rather than supplied as a trusted input, so the matched distance is bound to the committed template.

D. Threshold Check

The circuit enforces $d_H(\mathbf{p}, \mathbf{r}) \leq \tau$ where τ is a configurable threshold. The default configuration targets 50% similarity, corresponding to a maximum Hamming distance of:

$$\tau_{\text{default}} = 4096 \times (1 - 0.5) = 2048 \text{ bits}$$

The circuit performs the threshold comparison via bit decomposition and range checks. Stricter (60% similarity, $\tau = 1,638$) and relaxed (40% similarity, $\tau = 2,458$) configurations are also supported; in practice τ should be calibrated from data at the desired operating point (Section VI) rather than fixed a priori.

E. Circuit Parameters

The complete face verification circuit uses Halo2 with the SHPLONK (SHared PLONKish) polynomial commitment scheme over BN254. Table I summarizes the circuit configuration.

TABLE I
HALO2 CIRCUIT CONFIGURATION

Parameter	Value
Circuit rows (2^k)	$2^{16} = 65,536$
Advice columns	4
Lookup advice columns	1
Fixed columns	1
Lookup bits	13
Public inputs	5 (face, threshold, liveness, digest, commitment)
Polynomial commitment	KZG (SHPLONK)
Transcript hash	Blake2b-256
Scalar field	BN254 (~ 254 bits)

IV. SPATIAL FLASH LIVENESS PROTOCOL

Presentation attacks — where an adversary holds up a photo, video, or screen displaying the victim’s face — are a well-known threat to camera-based biometric systems. SABLE integrates a software-only liveness detection protocol based on controlled-illumination reflectance analysis, inspired by Tang et al. [1], and extends it with ZK proof integration and a cryptographic commit-reveal scheme.

A. Commit-Reveal Nonce Scheme

To prevent either party from predicting or manipulating the flash pattern, we use a two-phase commit-reveal protocol:

- 1) The client generates a 32-byte random nonce n_c and sends its SHA-256 commitment $H(n_c)$ to the server.
- 2) The server generates its own 32-byte random nonce n_s and sends it in cleartext.
- 3) The client reveals n_c ; the server verifies $H(n_c)$ to ensure the client committed honestly.
- 4) Both parties derive the flash color pattern deterministically from $\text{colors} = \text{HKDF-SHA256}(n_c || n_s)$.

Each challenge expires after 30 seconds and is single-use, preventing replay attacks and limiting the window for attack adaptation.

B. Split-Screen Color Flash

The derived HKDF output determines 6 RGB color values across 3 rounds, with each round displaying a *different* color on the upper and lower halves of the screen. Colors are subject to:

- **Photosensitive safety clamping** per WCAG 2.3.1, with luminance constrained to 40–100%.
- **Minimum angular distance** between paired colors, ensuring distinguishable differential reflectance.

C. 3D Geometry Detection

A real 3D face reflects split-screen illumination differently across regions due to facial geometry and surface normals; a flat photo or screen reflects both colors identically everywhere. For each flash round, the system captures the per-region response to the two different colors and computes delta vectors representing the change in RGB values from baseline. Spatial differentiation is confirmed when the cosine similarity between upper and lower delta vectors falls below a threshold,

indicating true 3D geometry rather than a flat presentation attack.

D. Delta Fingerprint Quantization

Per-region color responses are quantized into compact 16-bit delta fingerprints encoding the essential reflectance characteristics:

$$\underbrace{\text{order}}_{3\text{ b}} \mid \underbrace{\text{mid_ratio}}_{4\text{ b}} \mid \underbrace{\text{min_ratio}}_{4\text{ b}} \mid \underbrace{\text{magnitude}}_{5\text{ b}} \quad (3)$$

- **Order** (3 bits): Channel ordering (e.g., $R \geq G \geq B = 0$).
- **Mid ratio** (4 bits): Ratio of the middle channel to the maximum.
- **Min ratio** (4 bits): Ratio of the minimum channel to the maximum.
- **Magnitude** (5 bits): Overall brightness response (0–31).

E. ZK Proof of Liveness

The delta fingerprints are fed into the Halo2 circuit alongside the face-match check, producing a single composite proof. The circuit verifies three conditions per round:

- 1) **Color match:** $d_H(\text{fingerprint}_{\text{actual}}, \text{fingerprint}_{\text{expected}}) \leq \tau_c$ (the observed reflectance matches the expected pattern).
- 2) **Spatial differentiation:** $d_H(\text{fingerprint}_{\text{upper}}, \text{fingerprint}_{\text{lower}}) \geq \tau_s$ (upper and lower face regions responded differently, indicating 3D geometry).
- 3) **Magnitude check:** $\text{magnitude} \geq m_{\min}$ (adequate flash response was detected, ruling out non-reflective spoofs).

The verifier learns only pass/fail — no raw reflectance data, pixel values, or camera frames are exposed.

F. Verifiable Challenge Digest

A straightforward circuit design would treat the expected fingerprints ($\text{fp}_0, \dots, \text{fp}_5$) and thresholds ($\tau_c, \tau_s, m_{\min}$) as private witnesses. However, this creates a vulnerability: a malicious prover can substitute arbitrary values — for example, setting $\tau_c = 16$ to accept any fingerprint — while producing a cryptographically valid proof. The verifier would have no way to detect this substitution.

We address this by packing all nine liveness parameters into a single public input, exposed as a cryptographic commitment. The packing format (LSB-first) is:

$$\delta = \sum_{i=0}^5 \text{fp}_i \cdot 2^{16i} + \tau_c \cdot 2^{96} + \tau_s \cdot 2^{104} + m_{\min} \cdot 2^{112} \quad (4)$$

The total payload is 120 bits (six 16-bit fingerprints plus three 8-bit thresholds), which comfortably fits within the 254-bit BN254 scalar field without overflow.

Crucially, the circuit loads each threshold and expected-fingerprint witness exactly once and reuses the same assigned values for both the liveness checks and the digest packing. This constraint prevents a malicious prover from using different values in the checks versus the digest. The verifier independently

TABLE II
PERFORMANCE RESULTS (APPLE SILICON, RELEASE MODE)

Operation	Target	Achieved
Key generation + first proof (once)	—	~1.5 s
Proof generation (face + liveness)	≤ 1000 ms	~1.0 s
Verification	≤ 50 ms	~1.8 ms
Proof size	≤ 10 KB	2.08 KB
Spatial flash challenge	—	3 rounds, ~1 s

computes δ from the HKDF-derived nonces and verifies it against the public input, cryptographically binding the proof to the correct challenge parameters.

V. PERFORMANCE EVALUATION

We evaluate SABLE on an Apple M-series processor (ARM64) in release mode. Table II presents the key metrics.

Proving and verifying keys are generated once from a precomputed structured reference string and cached; including the first proof this one-time step takes approximately 1.5 s. Subsequent proofs generate in approximately 1 second, at the boundary of the interactive-authentication budget. This is larger than a threshold-only circuit because the matcher performs *all* of the matching in-circuit: it computes the Hamming distance over the two 512-byte embeddings, binds the enrolled template to its Poseidon commitment, and constrains every byte to be a valid thermometer code (Section VI), rather than accepting a precomputed distance as a trusted input. These additions raise the circuit from 2^{14} to 2^{16} rows but make the proof self-contained: a prover cannot assert an unverified distance or match against an unregistered template. Verification — succinct and independent of circuit size — completes in approximately 1.8 ms, effectively instantaneous, and the 2.08 KB proof size — constant in the embedding dimension — is well-suited for constrained channels such as NFC or BLE.

The complete end-to-end authentication flow, including three rounds of spatial flash (approximately 1 s), face capture, and proof generation, totals approximately 2–3 seconds, comparable to conventional smartphone biometric unlock. All figures are measured on a desktop-class Apple M4 (release mode, multi-threaded); on-device mobile performance has not yet been characterized (Section X).

The circuit uses $2^{16} = 65,536$ rows with 4 advice columns and KZG polynomial commitments. Performance benefits significantly from the lookup-table-based bit decomposition and XOR/popcount computation versus traditional field-arithmetic distance calculations.

VI. RECOGNITION ACCURACY AND MATCHER ENCODING

The bit-level Hamming matcher of Section III is efficient in-circuit, but under a *binary* 8-bit quantization it discards ordinal structure: adjacent quantization levels 127 (0111 1111) and 128 (1000 0000) differ in all eight bits. We quantify this accuracy cost and evaluate the drop-in thermometer encoding that SABLE deploys to remove it.

TABLE III
VERIFICATION EER (%) BY EMBEDDING BACKBONE AND MATCHER ENCODING. EACH ENCODING IS REPORTED UNDER ITS BEST PER-DIMENSION PRESCALE; THERMOMETER L9 USES THE SAME 8 BITS/DIM AS BINARY HAMMING. Δ IS THE BINARY-HAMMING PENALTY OVER FLOAT COSINE. N IS THE NUMBER OF USABLE PAIRS (FACE RES FAILS TO DETECT MANY CROSS-POSE CPLFW FACES). SAMPLED-PAIR PROTOCOL; RELATIVE GAPS ARE THE SIGNAL, NOT ABSOLUTE EERS.

Dataset	Backbone	N	Float	Binary	Δ	Thermo-L9
LFW	FaceRes	5998	4.63	7.97	+3.33	5.17
LFW	ArcFace	6000	4.33	4.47	+0.13	4.07
CALFW	FaceRes	5998	18.94	23.24	+4.30	19.07
CALFW	ArcFace	6000	6.47	6.80	+0.33	6.97
CPLFW	FaceRes	3004	26.04	29.48	+3.45	27.56
CPLFW	ArcFace	5854	6.33	8.02	+1.69	6.08

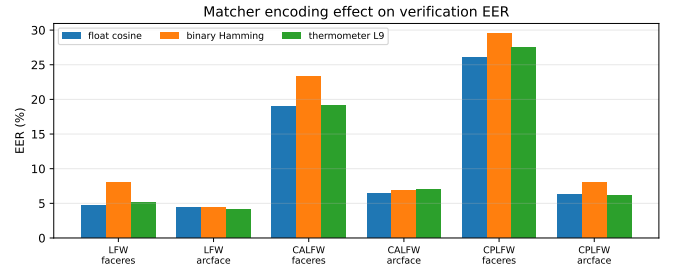


Fig. 1. Verification EER by backbone and matcher encoding. Binary Hamming inflates EER for the lightweight backbone across all three datasets; thermometer L9 (same circuit cost) recovers most of the gap. A strong backbone is barely affected.

A. Protocol

We extract embeddings with two backbones: the lightweight on-device model SABLE deploys (1024-d) and a strong reference model (ArcFace, 512-d), and evaluate verification on LFW and on its harder cross-age (CALFW) and cross-pose (CPLFW) variants. Each embedding is mapped into $[-1, 1]$ by a per-dimension prescale (each encoding is reported under its best prescale), then scored three ways: float cosine (the backbone ceiling), binary Hamming (Section III), and a *thermometer* encoding at nine levels. A thermometer code at L levels represents level k as k ones followed by zeros, so the Hamming distance between two codes equals the $L1$ distance between their levels — it restores ordinal structure. Crucially, $L = 9$ uses $L - 1 = 8$ bits per dimension, *the same width as binary*, so it reuses the Hamming circuit of Section III unchanged: the XOR and popcount work is identical, and the only added constraint is a per-byte check that the code is a legal thermometer pattern, realizable as a nine-entry lookup at negligible cost.

B. Results

Table III and Figure 1 show a consistent, backbone-dependent effect. For the lightweight deployed model, whose embedding components cluster near the 127/128 boundary, binary Hamming inflates EER by 3.3–4.3 points on every dataset, and degrades the low-false-accept operating region

most. The thermometer encoding recovers the large majority of this gap (e.g. 7.97% $\rightarrow$ 5.17% on LFW, 23.24% $\rightarrow$ 19.07% on CALFW) at no additional matching cost. For the strong backbone, whose embeddings are well spread, binary Hamming is nearly lossless (+0.1–1.7 points) and the encoding choice is immaterial. The practical implication is two independent levers: an ordinal in-circuit encoding, or a better-distributed embedding, each yields competitive accuracy under the ZK-friendly Hamming matcher.

Why tolerate a lossy matcher at all. The bit-level Hamming matcher is not chosen for accuracy but for in-circuit cost: the distance reduces to a per-bit XOR and a population count (Section III), both lookup-friendly and reusing the same bit-decomposition machinery as the liveness sub-circuit, which keeps the entire statement a small, uniform, transparent-setup circuit. A Euclidean or cosine matcher proved in-circuit would instead demand per-dimension squaring or normalized dot products in fixed-point arithmetic, enlarging the circuit and complicating the liveness integration. The results above show that the usual price of this efficiency—the ordinal information that binary encoding discards—is *not* intrinsic to the bit-level matcher: the thermometer encoding retains the identical XOR/popcount circuit (and the same 8 bits/dimension) yet recovers the large majority of the binary penalty. The bit-level matcher therefore need not be read as an accuracy-for-efficiency trade; with an ordinal code it is competitive with float cosine on the deployed backbone at no added in-circuit cost. SABLE adopts this thermometer encoding as its production matcher, with a stats-free tanh prescale as the zero-configuration default; the in-circuit matcher additionally constrains every byte to be a valid thermometer code, so the ordinal distance cannot be forged with a non-unary byte.

This evaluation also motivates calibrating the decision threshold from data rather than the fixed 50%-similarity default of Section III; the equal-error operating points we measure sit between 56% and 59% similarity for the binary matcher. The prescale statistics and decision threshold are deployment-time calibration parameters (like the threshold τ): a deployer fits them once on a representative set, analogous to a normalization layer’s running statistics.

C. Caveats

These experiments use a sampled-pair protocol over publicly mirrored images rather than the official fold definitions, so absolute EERs are not directly comparable to the literature (and the lightweight backbone is far from state-of-the-art); the *relative* gaps between encodings are the robust signal. The lightweight backbone additionally fails to detect many extreme-pose CPLFW faces, so its CPLFW figure is computed over a reduced, easier subset (reflected in the N column of Table III). A camera-ready evaluation should adopt the official verification protocols and add a dedicated presentation-attack study (Section IV) reporting APCER/BPCER.

A further caveat concerns the prescale. The thermometer (and binary) results are reported under each encoding’s best per-dimension prescale, and the z -score variant fits its

mean/std on the evaluation set itself — an unsupervised but *transductive* normalization. A deployment cannot use evaluation-set statistics; it must freeze calibration constants ahead of time, so the reported z -score recovery is a mild upper bound on what a deployment with frozen constants would achieve. The production default avoids this dependency entirely by using the stats-free tanh prescale, whose recovery should be re-measured under the official protocol; the ordinal structure that drives the improvement is prescale-independent, so thermometer is expected to retain its advantage over binary regardless.

VII. SECURITY ANALYSIS

A. Cryptographic Security

SABLE targets 128-bit security. The Halo2 proof system operates over BN254 (approximately 254-bit scalar field) with KZG polynomial commitments, providing computational soundness under the discrete logarithm and pairing assumptions. Pedersen commitments over BLS12-381 (381-bit base field, embedding degree 12) are computationally hiding and binding under the discrete logarithm assumption. Poseidon provides collision resistance over finite fields with parameters selected per the original security analysis [2].

Halo2’s transparent setup eliminates the trusted ceremony required by systems like Groth16, removing a significant trust assumption from the deployment model.

B. Replay Protection

Multiple mechanisms prevent replay attacks:

- **Challenge TTL:** Each liveness challenge expires after 30 seconds and is single-use, limiting the attack window.
- **Commit-reveal nonce protocol:** The client commits $H(n_c)$ before the server reveals n_s , preventing either party from unilaterally controlling the flash pattern.
- **HKDF-derived patterns:** Color patterns are deterministic yet unpredictable given the combined nonces, preventing pre-recorded response reuse.
- **Session nonce binding:** The proof encodes the challenge nonce, binding it to a specific authentication session.
- **Ephemeral session keys:** Peer-to-peer channels derive fresh session keys via X25519 ECDH with ephemeral key material.

C. Challenge Parameter Integrity

The challenge digest (Section IV-F) prevents a class of attacks where a malicious prover substitutes relaxed thresholds or arbitrary expected fingerprints while producing a cryptographically valid proof. Because the digest is a public input, the verifier independently computes the expected value from the HKDF-derived nonces and rejects proofs where the digest does not match. Furthermore, the circuit reuses the same witness cells for both the liveness checks and the digest computation, so there is no opportunity for the prover to use different values internally.

D. Presentation Attack Resistance

The spatial flash protocol raises the bar significantly beyond static attacks:

- **Photos:** A flat photo reflects split-screen colors identically across the surface, failing the spatial differentiation check.
- **Screen replay:** A screen displaying a face video also reflects illumination uniformly, as the display surface is flat.
- **Video manipulation:** The commit-reveal nonce scheme prevents an attacker from pre-computing the color sequence, as they cannot predict n_s before committing to n_c .

E. Honest Limitations

SABLE does not claim to defeat all presentation attacks:

- **3D masks:** A high-quality silicone or 3D-printed mask with realistic surface properties would reflect split-screen light similarly to a real face. Mitigation requires depth sensors or multi-spectral imaging, which SABLE explicitly avoids to maintain hardware independence.
- **Real-time video manipulation:** Sophisticated real-time deepfake systems that can modulate camera input in response to observed flash patterns are not addressed.
- **Quantum computing:** BLS12-381 and BN254 are vulnerable to Shor’s algorithm. Post-quantum migration (e.g., lattice-based commitments and proof systems) will be necessary as quantum computing matures.

VIII. RELATED WORK

Screen flash liveness. Tang et al. [1] demonstrated that controlled screen illumination can distinguish 3D faces from 2D spoofs by analyzing differential light reflectance. SABLE builds on this approach, extending it with a cryptographic commit-reveal nonce scheme for unpredictability, compact 16-bit delta fingerprint quantization for efficiency, and integration into a ZK proof circuit so that raw reflectance data is never exposed to the verifier.

ZK proof systems. Groth16 [3] provides succinct SNARK proofs but requires a per-circuit trusted setup ceremony. PLONK [4] introduced universal and updatable structured reference strings, reducing the trust burden. Halo2 [5] achieves transparent setup (no ceremony) with recursive proof composition, using KZG polynomial commitments. SABLE uses Halo2 with the SHPLONK variant for its combination of transparent setup, practical prover performance, and mature tooling.

Privacy-preserving biometrics. Homomorphic encryption approaches [6], [7] allow biometric matching on encrypted data but incur high computational costs and typically require server-side processing. MPC-based systems [8] split biometric templates across parties but require online coordination. FaceChain [9] and similar blockchain-based systems provide decentralized identity but rely on on-chain verification. Most recently, Mao et al. [11] proposed ZABA, which combines Pedersen vector commitments with multimodal cancelable

biometrics (fingerprint, face, iris) for e-health authentication, achieving ~ 78 ms proving time. However, ZABA requires an online authentication server, does not address liveness detection, and its Sigma-style proofs incur ~ 140 ms verification time — roughly $78\times$ slower than SABLE’s 1.8 ms succinct Halo2 verification. Lai et al. [10] proposed BioZero, which verifies biometric Euclidean distance on Ethereum via Groth16 proofs over homomorphic Pedersen commitments. Groth16 yields compact ~ 256 -byte proofs, but requires a per-circuit trusted setup ceremony; moreover, on-chain verification incurs gas costs and an internet dependency on every authentication. BioZero also lacks liveness detection. SABLE differs from all of the above by performing matching entirely in a ZK circuit on the user’s device with transparent setup, requiring no server-side computation, internet connectivity, or multi-party coordination, while additionally integrating liveness detection into the proof.

Anonymous credentials. BBS+ signatures [12], [13] enable selective disclosure of credential attributes. The W3C Verifiable Credentials standard provides an interoperability framework. SABLE’s architecture supports BBS+ credentials anchored to the biometric Pedersen commitment, enabling composite proofs that simultaneously attest biometric match and credential predicates.

IX. COMPARISON WITH HOMOMORPHIC ENCRYPTION

Homomorphic encryption (HE) is the most widely studied cryptographic technique for privacy-preserving biometrics. Yang et al. [14] provide a comprehensive survey of HE-based approaches across face, iris, fingerprint, gait, voice, and signature modalities. In HE-based systems, biometric templates are encrypted and matching is performed in the ciphertext domain — the server computes on encrypted data without ever observing the plaintext. SABLE takes a fundamentally different approach: matching is performed locally on the prover’s device, and only a succinct zero-knowledge proof of the result is transmitted to the verifier. This section compares the two paradigms across several dimensions.

A. Architectural Model

Homomorphic encryption positions the server as a *computing party*: it stores encrypted biometric templates and performs distance matching on ciphertexts, relying on the hardness of RLWE (for BFV/BGV/CKKS schemes) to protect the plaintext from disclosure. The server still processes biometric data, though in encrypted form.

SABLE uses a different architectural model: the server stores only a Pedersen commitment $C = g^h \cdot h_r^s$ (48 bytes), which is information-theoretically hiding, and acts purely as a *verifying party* that checks proofs. The server never processes biometric data in any form, which has significant practical implications. HE systems require server-side HE libraries and substantial compute resources for ciphertext operations, whereas SABLE’s verifier requires only a fast proof check (approximately 1.8 ms) with no specialized cryptographic libraries.

TABLE IV
COMPARISON WITH HE-BASED FACE VERIFICATION SYSTEMS

System	Match Time	Template Size	Scheme
Boddeti [7]	0.15 s	32 KB	BFV
Drozdzowski et al. [15]	0.4 s	65 KB	BFV
Pradel et al. [16]	47 s	512 KB	BGV
Sun et al. [17]	0.06 s	16 KB	CKKS
SABLE	0.25 s	2.08 KB	Halo2

B. Computational Efficiency

Table IV compares SABLE against representative HE-based face verification systems from the literature.

Recent surveys identify computational complexity as a primary HE challenge: “the computing time of some HE-based biometric systems could take more than 10 s,” and note that “ciphertext size typically exceeds plaintext size, so encrypted biometric data impose substantial computational and storage overhead” [14]. SABLE’s 2.08 KB proof is 8–250 times smaller than HE ciphertext templates, while ~ 1 s proof generation is in the range of HE matching times and additionally performs the match in-circuit and includes a cryptographic proof of liveness.

C. Privacy Strength

Both approaches satisfy the three ISO/IEC 24745 requirements for biometric template protection: irreversibility, unlinkability, and renewability. However, they differ in the degree of server-side exposure:

- **HE:** The server holds encrypted templates and computes encrypted distances. If the decryption key is compromised (the “fully compromised” scenario), all templates are exposed. Shahreza et al. [18] address this by combining cancelable biometrics with HE, but this adds implementation complexity.
- **SABLE:** The server holds only Pedersen commitments and never processes biometric data. Key compromise does not expose templates because no encrypted templates exist on the server. The zero-knowledge property ensures that even a malicious verifier learns only the boolean match result and the public threshold.

D. Liveness Detection

The HE literature typically treats liveness detection as orthogonal to template protection. Among recent systems surveyed, only Salem et al. [19] address liveness, and then only as a separate processing stage. Extending HE to liveness detection would require computing over encrypted reflectance data — operations like channel ordering, bit-level Hamming distance, and conditional logic that are expensive or infeasible with standard HE schemes (limited to addition and multiplication).

SABLE integrates liveness directly into the ZK circuit. Halo2’s arithmetic circuit naturally supports the bit decomposition, XOR-based Hamming distance, range checks, and conditional branches required for the spatial flash protocol,

adding approximately 3,200 constraints to the total circuit. This tight integration ensures cryptographic binding between liveness and face-match: a prover cannot produce a valid face-match proof without simultaneously proving liveness.

E. Operational Requirements

- **Server infrastructure:** HE matching requires server-side libraries (SEAL, HELib, OpenFHE) and substantial computational resources. SABLE requires only proof verification (approximately 1.8 ms) with no specialized cryptographic libraries.
- **Network connectivity:** HE systems require persistent client-server communication for encrypted queries and results. SABLE operates fully offline via peer-to-peer channels (NFC, BLE, WiFi Direct).
- **Hardware requirements:** HE systems require advanced specialized hardware platforms. SABLE runs on commodity mobile devices (Apple Silicon, Android) with no specialized biometric sensors.
- **Trusted setup:** HE requires key management infrastructure and public/private key escrow. SABLE uses Halo2’s transparent setup, eliminating the need for a trusted ceremony or centralized key authority.

F. Where HE Retains Advantages

HE-based approaches retain advantages in specific scenarios that SABLE’s architecture does not currently support:

- **1:N identification:** HE supports server-side search across encrypted templates to find matching identities. SABLE’s ZK proofs are inherently 1:1; the prover must know which enrolled commitment to verify against. Supporting server-side 1:N search over ZK commitments would require additional techniques such as private information retrieval.
- **Numerical precision:** The CKKS scheme natively supports approximate real arithmetic, preserving full embedding precision. SABLE’s ordinal quantization (nine thermometer levels per component) is coarser, which may affect decision boundaries at extreme similarity thresholds in specialized applications.
- **Server-side inference:** HE integrates naturally with neural network inference on encrypted data [17], [20], enabling privacy-preserving feature extraction server-side. SABLE requires feature extraction to occur on the user’s device before quantization.

G. Comparison with ZK-Based Biometric Systems

Two recent systems also apply zero-knowledge proofs to biometric authentication: ZABA [11] and BioZero [10]. Table V compares SABLE against both.

BioZero verifies Euclidean distance on-chain via Groth16 proofs over homomorphic Pedersen commitments. Groth16 produces the smallest proofs (approximately 256 bytes) but requires a per-circuit trusted setup ceremony; circuit modifications (adding liveness, adjusting embedding dimensions) necessitate repeating the ceremony. More significantly, reliance

TABLE V
COMPARISON WITH ZK-BASED BIOMETRIC AUTHENTICATION SYSTEMS

Property	SABLE	BioZero [10]	ZABA [11]
ZK scheme	Halo2 (PLONK)	Groth16	Sigma protocol
Trusted setup	No	Yes (per-circuit)	Yes (trusted entity)
Prove time	~250 ms	$O(s)^\dagger$	~78 ms
Verify time	~1.8 ms	On-chain [‡]	~140 ms
Proof size	2.08 KB	~256 B	$O(n)^\S$
Distance metric	Hamming	Euclidean	Euclidean
Modalities	Face	Face	Face+Finger+Iris
Liveness detection	Yes (ZK-proved)	No	No
Offline capable	Yes (P2P)	No (Ethereum)	No (server)
Verifier learns	Boolean result	Boolean result	Boolean result
Infrastructure	None	Blockchain node	Auth server
Post-quantum	No	No	No

[†] Groth16 proving is dominated by FFT and MSM; typically seconds for moderate circuits.

[‡] On-chain Groth16 verification costs ~200–300k gas on Ethereum.

[§] Sigma protocol transcripts grow linearly with witness size.

^{||} All three systems rely on elliptic curve assumptions vulnerable to Shor’s algorithm.

on Ethereum introduces gas costs and network latency on every authentication attempt. SABLE’s verification is purely local, incurs no costs, and completes in 1.8 ms without network communication.

ZABA uses Pedersen commitments with Sigma-protocol proofs supporting multiple modalities (fingerprint, face, iris). While ZABA’s proving time (approximately 78 ms) is fast, verification at approximately 140 ms is roughly 78 times slower than SABLE’s 1.8 ms. Sigma protocol transcripts scale linearly with witness size, whereas SABLE’s PLONK proofs remain succinct (2.08 KB constant size). ZABA requires an online authentication server and a centralized trusted authority for parameter generation.

Neither BioZero nor ZABA integrates liveness detection. Adding liveness to BioZero requires modifying the circuit and rerunning the trusted setup; extending ZABA to liveness would require supporting non-algebraic operations (bit decomposition, channel ordering, conditionals) that Sigma protocols do not naturally express. SABLE’s Halo2 circuits accommodate liveness for approximately 3,200 additional constraints within the same transparent-setup framework.

Recognition accuracy. The privacy layer in HE, BioZero, and ZABA is *accuracy-transparent*: encrypted (CKKS) or commitment-based matching evaluates the same real-valued or Euclidean distance as the plaintext backbone, so recognition accuracy is inherited unchanged. SABLE’s bit-level matcher is the exception—its quantized Hamming distance is lossy—which is the cost of keeping matching inside a single transparent-setup circuit that also proves liveness. Section VI shows this cost is largely recoverable in place: the thermometer encoding restores ordinal structure within the unchanged Hamming circuit, narrowing the gap to float cosine to within roughly one EER point on the deployed backbone at no added matching cost. SABLE thus approaches the accuracy-transparency of the algebraic-distance systems without adopting their per-circuit ceremony (BioZero) or online server (ZABA).

In summary, HE and ZK proofs represent complementary privacy approaches. HE suits scenarios requiring server-side encrypted computation, while ZK suits client-side verification where biometric data must never leave the user’s device. Among ZK systems, SABLE’s distinguishing features are transparent setup, succinct proofs, integrated liveness, and offline operation. The primary limitation is the lack of 1:N identification, where HE and BioZero have structural advantages for database search scenarios.

X. CONCLUSION

We have presented SABLE, a privacy-preserving biometric authentication system combining Halo2 zero-knowledge proofs, Pedersen commitments, Poseidon hashing, and spatial flash liveness detection into a cohesive, offline-capable architecture. SABLE generates combined face-match and liveness proofs — computing the match, template-commitment binding, and liveness check entirely in-circuit — in approximately 1 second with 2.08 KB proof size, operates fully peer-to-peer without internet or specialized hardware, and supports selective attribute disclosure via BBS+ signatures. The spatial flash protocol resists photo and video replay attacks through 3D geometry differentiation, while the commit-reveal nonce scheme cryptographically prevents pattern prediction and manipulation.

SABLE is released as open-source under the Apache 2.0 license. Future directions include post-quantum cryptographic migration, formal circuit constraint verification, multi-spectral liveness detection, and comprehensive real-device validation across diverse hardware and lighting environments.

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